



Research article

Towards a modelling, optimization and predictive control framework for smart irrigation

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ABSTRACT

Smart irrigation scheduling is a promising approach for improving the efficiency and sustainability of agricultural water use, especially in arid and semi-arid lands. In this paper, we present a design and simulation of a model predictive controller for smart irrigation scheduling. The proposed controller is based on a mathematical model of the irrigation system, which is used to predict the future states of the system and determine the optimal irrigation schedule for a given crop and field. This study further evaluates the impact of the predictive control framework on crop yield, water use efficiency (WUE), and water savings in tomato production. Three control strategies, including manual control, open-loop control, and model predictive control (MPC), were compared using a Completely Randomized Design (CRD) methodology. Results indicate that MPC outperforms the other strategies, yielding the highest average crop yield of 20 t/ha and demonstrating superior WUE at 10.4 kg/m³. Additionally, MPC significantly reduces water consumption, achieving a 29 % and 8 % savings compared to manual and open-loop control, respectively. These findings underscore the efficacy of MPC in optimizing crop yield, conserving water resources, and promoting sustainable agriculture practices. The proposed controller has the potential to address the global water scarcity challenge and contribute to the sustainability of agriculture in arid and semi-arid lands.

1. Introduction

Water is a finite and precious resource, vital for sustaining life and supporting agriculture, the backbone of global food production. As the world's population continues to grow and climate change exacerbates water scarcity issues, efficient water management in agriculture becomes an imperative [1–4]. Smart irrigation, characterized by the integration of advanced modeling, optimization, and predictive control techniques, offers a transformative approach to address these challenges [5–7].

Smart irrigation systems have been developed as a promising solution for improving water use efficiency in agriculture [8–13]. These systems use a network of sensors and automation technology to monitor soil moisture, weather conditions, and crop water requirements, and adjust irrigation scheduling accordingly [14–16]. Smart irrigation systems have been shown to reduce water

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consumption by up to 30–50 % while maintaining or even increasing crop yields [17,18].

Existing irrigation scheduling methods typically rely on fixed schedules or simple threshold-based rules, which may not be optimal under varying conditions [19,20]. For example, a fixed schedule may result in over-irrigation during wet periods, while under-irrigation during dry periods. Threshold-based rules may not consider the complex interactions between soil moisture, weather conditions, and crop growth stages. This can result in reduced crop yields, water waste, and negative environmental impacts.

Irrigation control is critical in the implementation of smart irrigation systems [6]. It either takes the form of an open-loop system or a closed-loop system [21–24]. Whereas an open-loop system depends on the heuristic rules to decide on the quantity and timing of irrigation, a closed-loop irrigation control strategy relies on a plant model to be able to predict future outputs of the state variables [19, 25,26]. Over the years, conventional irrigation systems have relied on open-loop irrigation control strategies in scheduling irrigation and the results have shown reduced water use efficiency due to either over-irrigation or under-irrigation. Under open-loop irrigation control, automation of the irrigation system (sprinkler, drip, or surface irrigation) is either volume-based or time-based. This has made irrigation efficient though it still requires farmer intervention and does not adjust to real-time plant conditions. Closed-loop irrigation control is further subdivided into intelligent, linear, and optimal control [21,23,24].

Model predictive control is an optimal control technique that uses a mathematical model of the system behavior to predict the future states of the system and optimize the control inputs in real-time [27–29]. Model predictive control has been applied in various fields, including process control, automotive control, and building control [21,30–33]. In recent years, model predictive control has also been applied in irrigation scheduling and has shown promising results in improving water use efficiency and crop yield. Model predictive control has been shown to outperform existing irrigation scheduling methods, particularly under variable weather conditions.

Several studies have proposed model predictive controllers for irrigation scheduling using different mathematical models and optimization algorithms [19,34–40]. For example, Lozoya et al. [42] used sensor-based model predictive control for closed-loop irrigation. Delgoda et al. [43] implemented model predictive control to optimize irrigation amounts by minimizing root zone soil

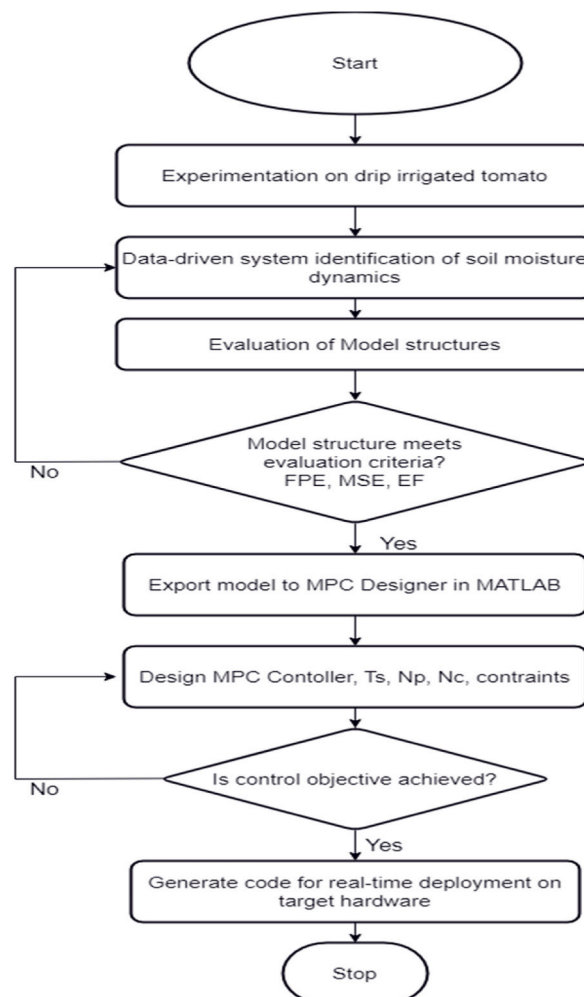


Fig. 1. Model predictive control design process.

moisture deficit. Recently, Abioye et al. [44] deployed a model predictive control algorithm on a Raspberry pi for precision irrigation in a controlled environment. Chen et al. [45] developed a model predictive control (MPC) framework for regulating stem water potential. However, these studies have mainly focused on specific crops, irrigation systems, or controlled environment agricultural systems. There is a need for a more generalized and adaptable model predictive controller that can be applied to different crops and open field conditions. The proposed study aims to address this gap in the literature by developing a model predictive controller for smart irrigation scheduling that is adaptable to different crops and field conditions.

This study aimed to design a model predictive controller for smart irrigation scheduling. The specific objectives were; (i) to minimize a performance index that represents a trade-off between a set of conflicting objectives, such as optimizing control inputs and satisfying constraints. (ii) to predict the future behavior of the process using a model and use the predictions to make control decisions. (iii) to handle constraints on inputs, outputs, and states, such as saturation limits, maximum allowed changes in inputs, and safety limits. (iv) provide robustness to disturbances and model uncertainties. (v) handle multivariable and nonlinear systems with coupling between inputs and outputs.

2. Materials and methods

2.1. System identification of soil moisture dynamics

The flowchart presented in Fig. 1 shows the approach used in designing the model predictive controller. First, experimentation on drip irrigated tomato was conducted and sensor data collected. The data was then used in system identification toolbox in MATLAB to identify a model structure which was later exported to the model predictive control designer app in MALAB. Tuning of parameters was done until the control objective was achieved.

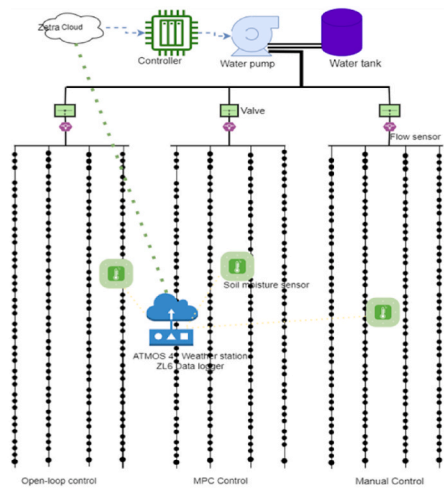
A data-driven experiment was conducted at the University for Development Studies in Northern Ghana as shown in Fig. 2. The experiment involved sensing weather parameters using an ATMOS 41 Weather station, TERROS 12 soil moisture sensor and YFS 201 flow sensor for irrigation volume measurement.

2.2. State space formulation of the control problem

A total number of 9674 experimental results was obtained from the data collection phase. This was split into two; 70 % of the data was used for model estimation while 30 % was used for model validation as proposed by Ref. [46]. The data collection followed the soil agro-hydrological model (Fig. 3) with Evapotranspiration, Rainfall and Irrigation volume as inputs and Soil moisture as Output. This is a multi-input single-output system (MISO) with evapotranspiration, irrigation and precipitation as inputs and soil moisture as the single output. Model predictive control is capable of handling both MIMO and MISO systems [26]. Different model structures were evaluated [17], and the resulting representation was a state space model as shown in Equation 1(a-g) [17].

$$\mathbf{x}(t+Ts) = \mathbf{A} \mathbf{x}(t) + \mathbf{B} \mathbf{u}(t) + \mathbf{K} \mathbf{e}(t) \quad (1a)$$

$$\mathbf{y}(t) = \mathbf{C} \mathbf{x}(t) + \mathbf{D} \mathbf{u}(t) + \mathbf{e}(t) \quad (1b)$$



(a)



(b)

Fig. 2. (a) Field layout of different irrigation control strategies, (b) Experimental field for data collection.

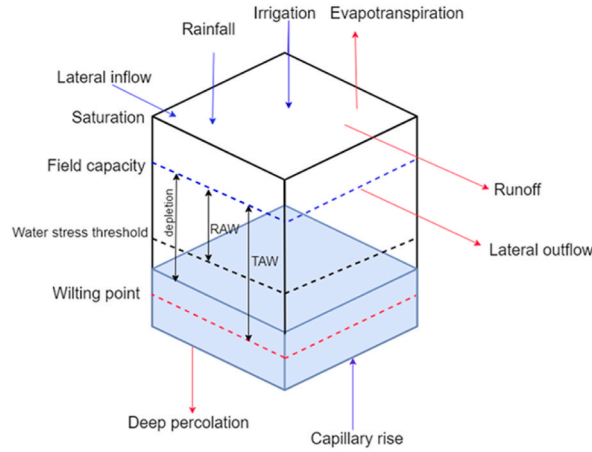


Fig. 3. Agro-hydrological model.

$$A = \begin{matrix} & \begin{matrix} x1 & x2 \end{matrix} \\ \begin{matrix} x1 \\ x2 \end{matrix} & \begin{bmatrix} 1 & -0.0008092 \\ -0.01197 & 0.9167 \end{bmatrix} \end{matrix} \quad (1c)$$

$$B = \begin{matrix} & \begin{matrix} u1 & u2 & u3 \end{matrix} \\ \begin{matrix} x1 \\ x2 \end{matrix} & \begin{bmatrix} -7.977e-07 & 7.204e-07 & -1.457e-05 \\ -0.0001524 & 9.12e-05 & 0.001182 \end{bmatrix} \end{matrix} \quad (1d)$$

$$C = \begin{matrix} & \begin{matrix} x1 & x2 \end{matrix} \\ \begin{matrix} y1 \end{matrix} & \begin{bmatrix} -22.49 & 0.0009123 \end{bmatrix} \end{matrix} \quad (1e)$$

$$D = \begin{matrix} & \begin{matrix} u1 & u2 & u3 \end{matrix} \\ \begin{matrix} x1 \end{matrix} & \begin{bmatrix} 0 & 0 & 0 \end{bmatrix} \end{matrix} \quad (1f)$$

$$K = \begin{matrix} & \begin{matrix} y1 \end{matrix} \\ \begin{matrix} x1 \\ x2 \end{matrix} & \begin{bmatrix} -0.0463 \\ -0.005392 \end{bmatrix} \end{matrix} \quad (1g)$$

Where; $x(t)$ represents the soil moisture, $u(t)$ represents the inputs to the system, $y(t)$ represents the response of the system, $e(t)$ represents disturbances, A, B, C, and D are input and out matrices.

The resulting model has an estimated fit of 98.92 %, Final Prediction Error of 1.856×10^{-07} and Mean Square Error of 1.851×10^{-07} .

2.3. Proposed smart irrigation control technique

Based on the state space equation the process dynamics model in the previous section, a model predictive controller can then be designed. This was designed basing on Equations 2–17 presented in Rawlings et al. [26]:

$$x_m(k+1) = A_m x_m(k) + B_m u(k) + B_d \omega(k) \quad (\text{Equation 2})$$

$$y(k) = C_m x_m(k) \quad (\text{Equation 3})$$

Where;

$\omega(k)$ = Input disturbance.

$\omega(k)$ is assumed to be a sequence of intergarted white noise, this means white noise sequence $\epsilon(k)$ by the difference equation.

$$\omega(k) - \omega(k-1) = \epsilon(k) \quad (\text{Equation 4})$$

Note that from Equation (2), it can be deduced that

$$x_m(k) = A_m x_m(k-1) + B_m u(k-1) + B_d \omega(k-1) \quad (\text{Equation 5})$$

Be defining $\Delta x_m(k) = x_m(k) - x_m(k-1)$ and $\Delta u(k) = u(k) - u(k-1)$ then subtratcting Equation (2) from Equation (5) gives Equation 6

$$\Delta x_m(k+1) = A_m \Delta x_m(k) + B_m \Delta u(k) + B_d \epsilon(k) \quad (\text{Equation 6})$$

In order to relate the output $y(k)$ to the state variable $\Delta x_m(k)$, we deduce that

$$\Delta y(k+1) = C_m \Delta x_m(k+1) = C_m(A_m \Delta x_m(k) + B_m \Delta u(k) + B_d \epsilon(k)) = C_m A_m \Delta x_m(k) + C_m B_m \Delta u(k) + C_m B_d \epsilon(k) \quad (\text{Equation 6})$$

Where, $\Delta y(k+1) = y(k+1) - y(k)$.

Choosing a new state variable vector $x(k) = [\Delta x_m(k)^T \ y(k)^T]^T$, we have Equations (7) and (8).

$$\begin{bmatrix} \Delta x_m(k+1) \\ y(k+1) \end{bmatrix} = \begin{bmatrix} A_m & 0_m^T \\ C_m A_m & I_{q \times q} \end{bmatrix} \begin{bmatrix} \Delta x_m(k) \\ y(k) \end{bmatrix} + \begin{bmatrix} B_m \\ C_m B_m \end{bmatrix} \Delta u(k) + \begin{bmatrix} B_d \\ C_m B_d \end{bmatrix} \epsilon(k) \quad (\text{Equation 7})$$

$$y(k) = \begin{bmatrix} O_m & I_{q \times q} \end{bmatrix} \begin{bmatrix} \Delta x_m(k) \\ y(k) \end{bmatrix} \quad (\text{Equation 8})$$

Where $I_{q \times q}$ is the identity matrix with dimensions $q \times q$ which is the number of outputs; and O_m is a $q \times n_1$ zero matrix. A_m , B_m , and C_m have dimension $n_1 \times n_1$, $n_1 \times m$ and $q \times n_1$, respectively.

2.3.1. Eigenvalues of the augmented model

The characteristic polynomial equation of the augmented model is given in Equation (9).

$$\rho(\lambda) = \det \begin{bmatrix} \lambda I - A_m & O_m^T \\ -C_m A_m & (\lambda - 1)I_{q \times q} \end{bmatrix} = (\lambda - 1)^q \det(\lambda I - A_m) = 0 \quad (\text{Equation 9})$$

2.3.2. Controllability and observability of the augmented model

In order for a predictive control system to attain the desired closed-loop control performance, controllability must be ensured. Similarly, a successful design of an observer is contingent on observability. It is possible to relax the conditions if closed-loop stability is a matter of concern.

2.4. Predictive control of the MISO system

In a MISO environment, the input and output are defined as shown in Equations (10) and (11).

$$\Delta U = [\Delta u(k_i)^T \ \Delta u(k_i + 1)^T \ \dots \ \Delta u(k_i + N_c - 1)^T]^T \quad (\text{Equation 10})$$

And the predicted output given as in Equation 11

$$Y = [y(k_i + 1 | k_i)^T \ y(k_i + 2 | k_i)^T \ \dots \ y(k_i + N_c | k_i)^T]^T \quad (\text{Equation 11})$$

These can be written as shown in Equation (12).

$$Y = Fx(k_i) + \Phi \Delta U \quad (\text{Equation 12})$$

$$\text{Where } F = \begin{bmatrix} CA \\ CA^2 \\ CA^3 \\ \vdots \\ CA^{N_p} \end{bmatrix}; \Phi = \begin{bmatrix} CB & 0 & 0 & \dots & 0 \\ CAB & CB & 0 & \dots & 0 \\ CA^2B & CAB & CB & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ CA^{N_p-1}B & CA^{N_p-2}B & CA^{N_p-3}B & \dots & CA^{N_p-N_c}B \end{bmatrix}.$$

The incremental optimal control within one optimization window is given by Equation 13

$$\Delta U = (\Phi^T \Phi + \bar{R})^{-1} (\Phi^T \bar{R}_s r(k_i) - \Phi^T Fx(k_i)) \quad (\text{Equation 13})$$

Where;

Matrix $\Phi^T \Phi$ has dimension $mN_c \times mN_c$ and $\Phi^T F$ has dimension $mN_c \times n$, and $\Phi^T \bar{R}_s$ equals the last q columns of $\Phi^T F$. The weight matrix $\bar{R}$ is a block matrix with m blocks and has its dimensions equal to the dimensions of $\Phi^T \Phi$.

The set point signal is $r(k_i) = [r_1(k_i) \ r_2(k_i) \ \dots \ r_q(k_i)]^T$ as the set-point signals to the single-output system.

Applying the receding horizon control principle, the first m elements in ΔU are taken to form the incremental optimal control in Equation (14).

$$\Delta u(k_i) = \overbrace{[I_m \ 0_m \ \dots \ 0_m]}^{N_c} (\Phi^T \Phi + \bar{R})^{-1} (\Phi^T \bar{R}_s r(k_i) - \Phi^T Fx(k_i)) = K_y r(k_i) - K_{mpc} x_i \quad (\text{Equation 14})$$

Where;

I_m and 0_m are respectively, the identity and zero matrix with dimension $m \times m$.

Fig. 4 shows the block diagram of model predictive controller for smart irrigation scheduling.

The cost function (J) for the error minimization control is calculated by a weighted square average of the control error and change manipulated variable of the optimization is formulated as shown in Equation (15).

$$\min_{\Delta u} J = (R_f - Y)^T Q_s (R_f - Y) + \Delta u^T Q_i \Delta u \quad (\text{Equation 15})$$

The target soil moisture values are denoted by R_f , where Y represents the future value of soil moisture and Q is a function that takes into account the state variable x and the input variable u . Tuning Q_s and Q_i can improve the error and control tracking performance of the model predictive controller. The cost function is subject to constraints (Equations (16) and (17)) such that the initial state is measured and maintained between the field capacity and permanent wilting point of the soil. Additionally, the minimized cost function must adhere to the system dynamics and the irrigation volume IR_{max} , in order to prevent over-irrigation.

$$0 \leq u(k) \leq IR_{max} \quad (\text{Equation 16})$$

$$SM_{PWP} \leq \theta \leq SM_{FC} \quad (\text{Equation 17})$$

Where IR_{max} is the effective irrigation, SM_{PWP} is the soil moisture at permanent wilting point and SM_{FC} is the soil moisture at field capacity. The optimal control signal is obtained as follows in Equation (18) and (19),

$$\frac{\partial J}{\partial \Delta U} = 0 \quad (\text{Equation 18})$$

$$\Delta U = (Q^T \Phi + \bar{R})^{-1} \Phi^T (R_f - \varphi x(k)) \quad (\text{Equation 19})$$

Tables 1 and 2 present the parameters used to design the model predictive controller and model predictive algorithm respectively.

3. Results and discussions

3.1. Data-driven experimentation

Figs. 5 and 6 present the data collected during the study including precipitation, reference evapotranspiration, soil moisture and irrigation amounts. The initial soil moisture content recorded in Fig. 6 was $0.34 \text{ m}^3/\text{m}^3$ at the beginning of the experiment. This coincided with the end of the rainy season, indicating high precipitation levels during that time. Subsequently, as the precipitation decreased, the soil moisture content gradually reached field capacity and was maintained through irrigation. The dynamics of the soil moisture content underwent changes due to factors such as plant water uptake and environmental conditions (weather), which led to water loss from the soil. Consequently, the volumetric water content exhibited varying patterns. The process of evapotranspiration further contributed to water loss, causing fluctuations in the soil's volumetric water content. These changes indicated the need for adjustments in irrigation, either increasing or reducing the amount of water provided. Hence, these factors directly influenced the irrigation requirements, as the amount of water applied to the plants was influenced by the varying variables, including plant water absorption, environmental effects, and evapotranspiration.

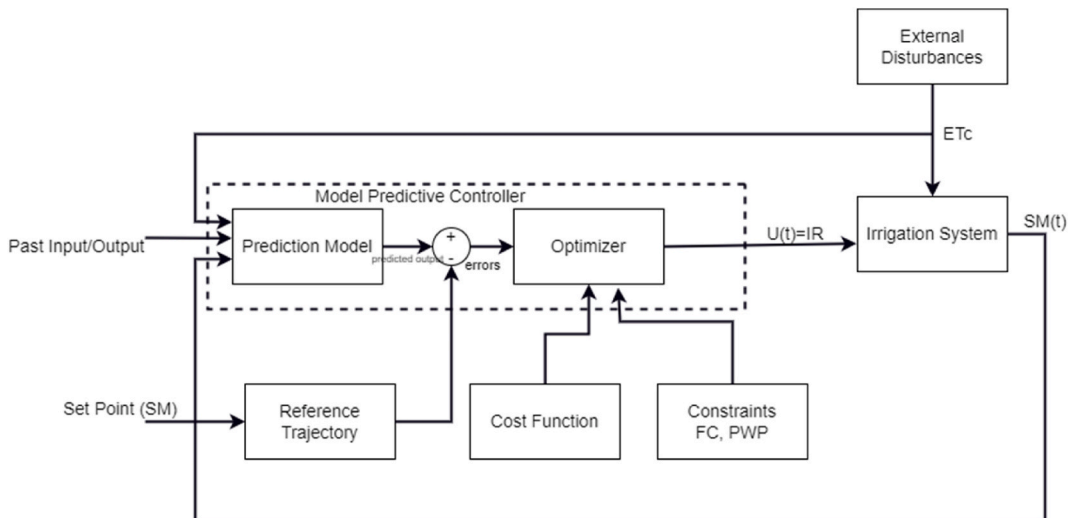


Fig. 4. Block diagram of a model predictive controller for smart irrigation.

Table 1

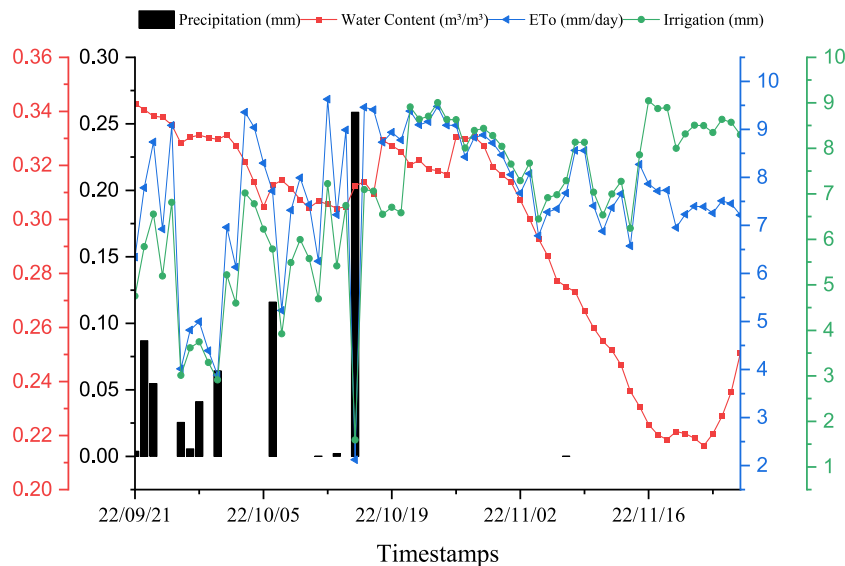
Parameters used to design the model predictive controller.

Parameters	Description	Value	Unit
T_s	Sampling Time	10	minutes
N_c	Control Horizon	2	–
N_p	Prediction Horizon	10	–
SM_{PWP}	Permanent Wilting Point	0.12	m^3/m^3
SM_{FC}	Field Capacity	0.25	m^3/m^3

Table 2

MPC algorithm for smart irrigation.

Algorithm 1
Required: Input $IR[k_i - 1]$, $ETc[k_i - 1]$, $P[k_i - 1]$
Output $\theta[k_i - 1]$
Use the previous input and output data to identify a predictive model
Loop
Read $\theta[k_i]$ #Current process output
Predict the process output $\theta_p[k_i]$ over a horizon
Estimate the error $e[k_i]$ and minimize $ J $
If $\theta_p[k_i] - \theta[k_i] > 0$ or $ETc \geq ETc_{max}$
Generate $u(k)$, $k = 1, \dots, m$
else
wait for the next sampling time $k = k + 1$
End

**Fig. 5.** A graph of ET_0 , water content, Irrigation and Precipitation during the experimental period [17].

3.2. Data-driven model predictive control in SIMULINK

The model predictive controller block was constructed through the implementation of a state space model that was identified using data-driven system identification, and then exported into the MPC Designer application for configuration and tuning. The performance of the predictive controller is largely dependent on the choice of the control and prediction horizons, as well as the constraints placed on the control variable. Following the design process, the controller was saved as a mat file in the MATLAB workspace, and subsequently integrated with the event generator function in Simulink (as seen in Fig. 7). To optimize the performance of the controller, an event-based generator function was integrated to update the designed controller during the next triggering event. The event generator function incorporates a decision function that determines the next triggering event time and generates a control action based on two conditions.

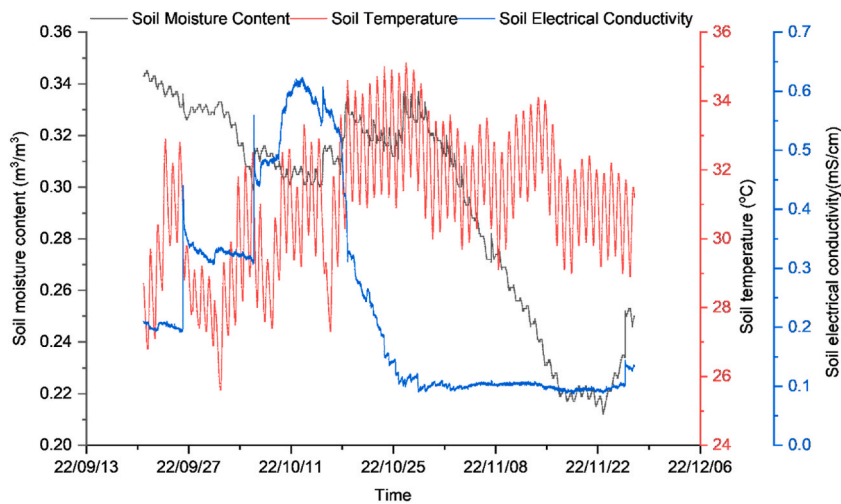


Fig. 6. Terros 12 soil moisture, temperature and electrical conductivity measurements [17].

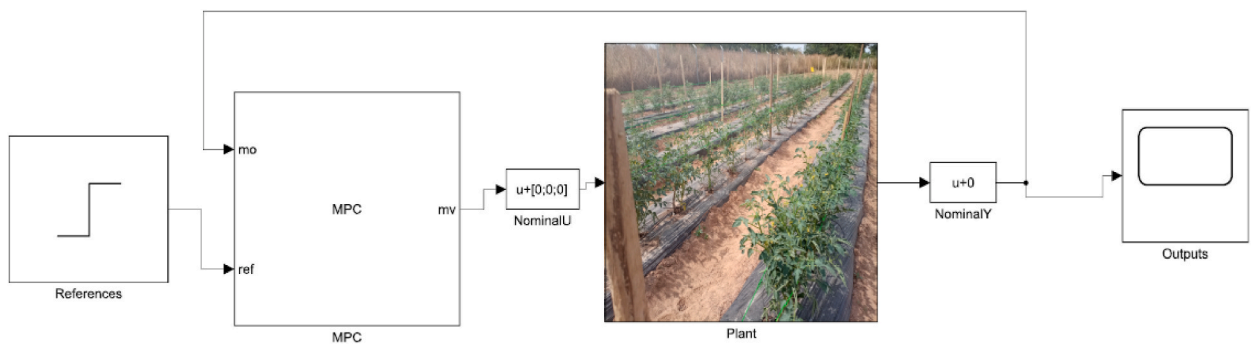


Fig. 7. Schematic of model predictive control in Simulink.

3.3. Input and output response

Figs. 8 and 9 present the input and out response of the model. The sensitivity analysis showed that the performance of the proposed controller was most sensitive to the prediction horizon and the soil moisture threshold. The longer the prediction horizon, the better the performance of the controller, as it was able to capture the dynamics of the system more accurately.

The sensitivity analysis showed that the performance of the proposed controller was most sensitive to the prediction horizon and the soil moisture threshold. The longer the prediction horizon, the better the performance of the controller, as it was able to capture the dynamics of the system more accurately. However, a longer prediction horizon also increased the computational complexity of the controller. The sensitivity analysis also showed that the controller was most effective when the soil moisture threshold was set to a moderate level, allowing for some degree of soil moisture depletion before irrigation.

Fig. 10 shows how the system's soil moisture level changes over time in response to changes in the input variables. The x-axis represents time, and the y-axis represents soil moisture level. Each input variable is plotted as a separate line on the graph, with different colors representing each input.

The input response plot helps to identify the sensitivity of the system to changes in the input variables. For example, if the plot shows that the soil moisture level is very sensitive to changes in irrigation, then it may be important to control irrigation very precisely in order to maintain the desired soil moisture level.

Furthermore, the input response plot can be used to help tune the parameters of the model predictive controller (MPC). By observing the plot, the MPC designer can adjust the parameters of the controller to achieve the desired response to changes in the input variables. This can help to ensure that the system maintains the desired soil moisture level, while minimizing the amount of water used for irrigation.

3.4. Experimental results

To determine the effect of the control framework on yield and water use efficiency on a tomato production, an experiment was

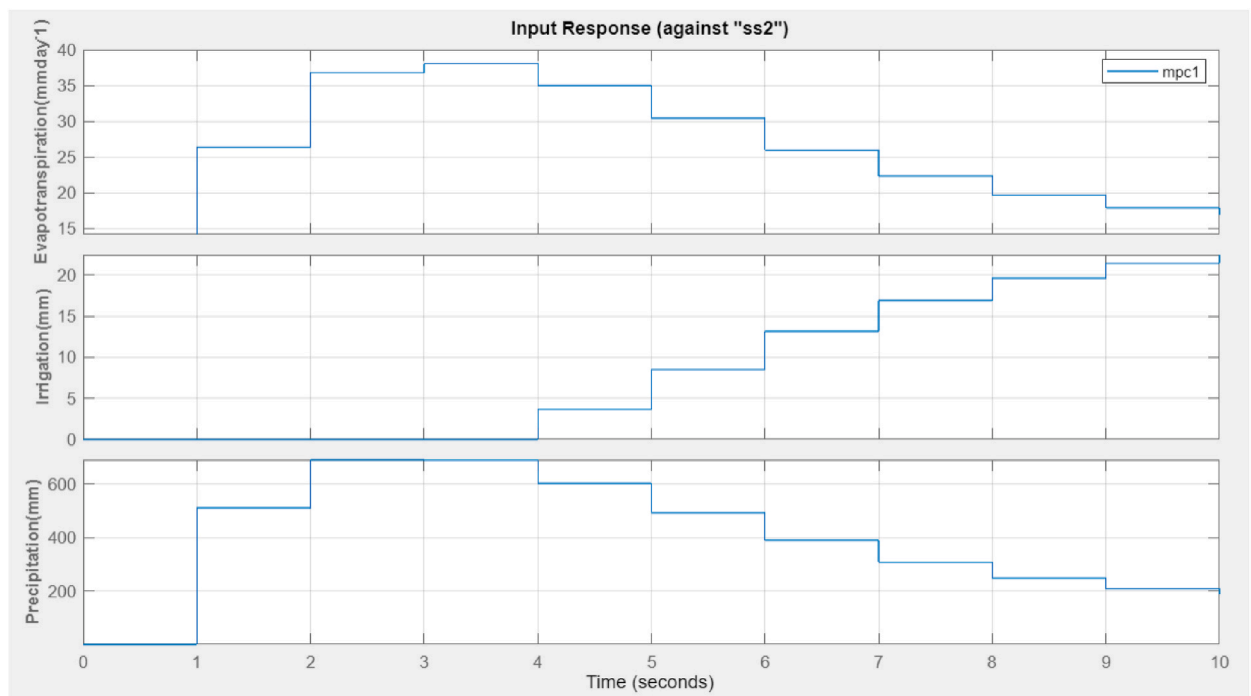


Fig. 8. Input response against the plant model.

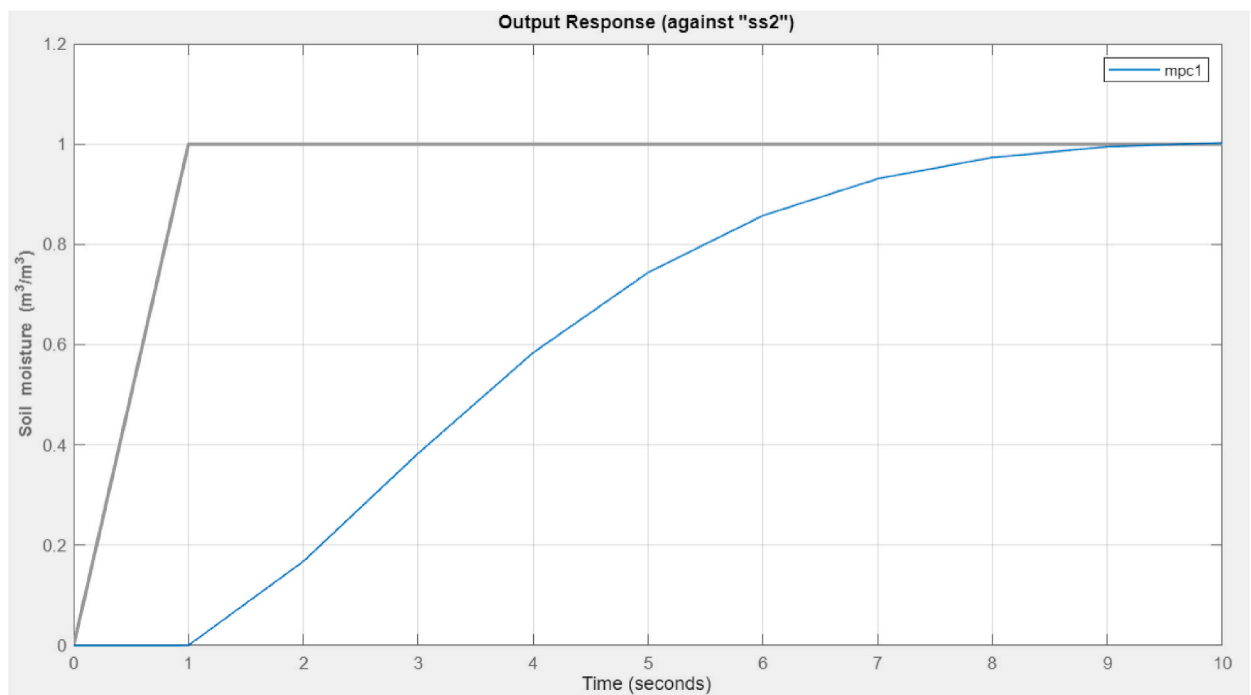


Fig. 9. Output response against the plant model.

designed. Employing a one-way Completely Randomized Design (CRD), the study incorporates three distinct treatments: Model Predictive Control (MPC) control, Open-loop control, and Manual control (Fig. 2). Each treatment was replicated four times to enhance result reliability. Utilizing a CRD ensures a fair comparison of treatments while mitigating potential external factors' influence on outcomes. Through the comparison of MPC control, Open-loop control, and manual control, this research offers valuable insights into

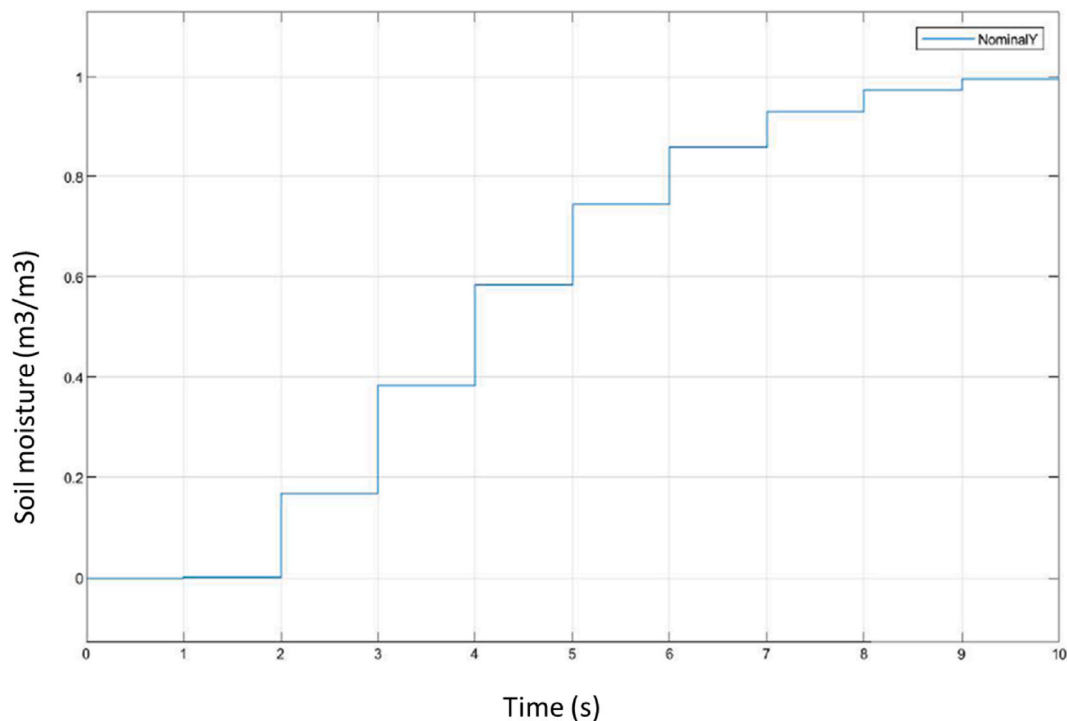


Fig. 10. MPC Simulation of soil moisture dynamics.

the efficacy of model-based smart irrigation control systems in optimizing water usage efficiency within tomato cultivation. The results are presented in Fig. 11a and b and c.

Results show that model predictive control (MPC) yields the highest average yield at 20 t/ha, followed by open-loop control at 16 t/ha and manual control at 14 t/ha. This signifies the significant influence of control strategy on crop yield, with MPC being the most effective due to its consideration of various parameters like environmental conditions and crop characteristics. On the other hand, manual control, solely reliant on the farmer's judgment, proves the least effective.

In terms of water use efficiency, MPC again emerges as the most effective strategy, with an average of 10.4 kg/m³, followed by open-loop control at 7.1 kg/m³ and manual control at 5.6 kg/m³. MPC's superior performance is attributed to its comprehensive consideration of environmental factors and crop conditions, allowing for precise water input predictions. Manual control, being judgment-based, demonstrates the lowest water use efficiency.

Furthermore, the implementation of MPC control leads to significant water savings, with a cumulative weekly consumption of 25,200 L compared to 32,400 L under manual control and 27,200 L under open-loop control. The consistent trend of MPC resulting in lower water consumption over time highlights its adaptability to changing crop water requirements. These findings underscore the importance of employing model-based control strategies like MPC to enhance water use efficiency and promote sustainable crop production.

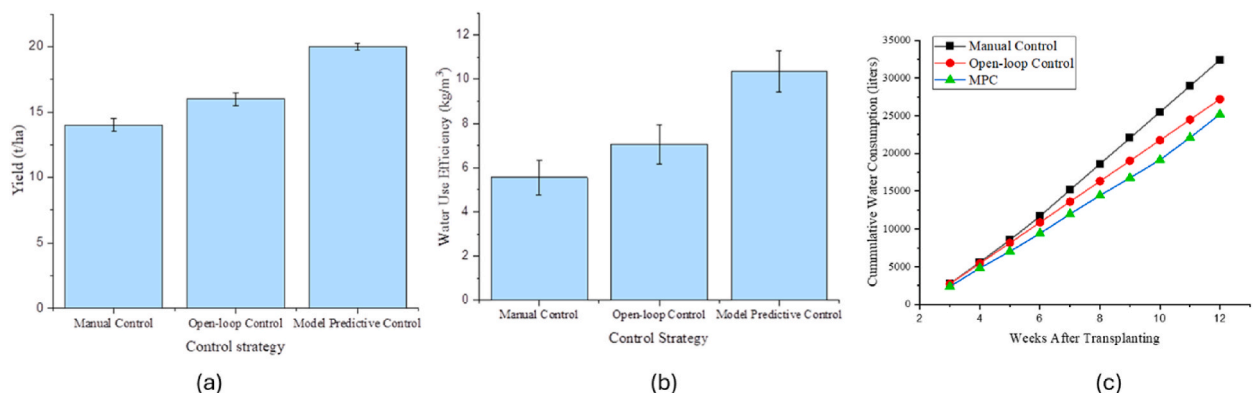


Fig. 11. Effect of predictive control framework on (a) yield, (b) water use efficiency and (c) water saving.

4. Challenges and limitations of the proposed framework

Complexity: The implementation of the model predictive control (MPC) framework in irrigation scheduling presents a notable challenge due to its inherent complexity [41]. This complexity stems from the requirement for sophisticated mathematical models capable of accurately predicting crop water requirements. Developing and calibrating these models necessitate extensive data on various factors such as crop growth dynamics, soil properties, weather conditions, and the characteristics of the irrigation system. Consequently, navigating through this complexity can be time-consuming and demanding, posing a significant obstacle to the widespread adoption of MPC in agricultural practices.

Data Requirements: A critical limitation of the MPC framework lies in its heavy reliance on real-time data inputs [47–49]. These inputs encompass a wide range of parameters including weather forecasts, soil moisture levels, and crop conditions. However, ensuring the availability and accuracy of these data sources can be challenging, particularly in agricultural regions that lack adequate infrastructure for data collection and dissemination. As a result, the effectiveness of MPC in optimizing irrigation schedules may be compromised by the unavailability or inaccuracy of essential data inputs.

Model Uncertainty: Despite efforts to develop robust mathematical models, uncertainty remains inherent in predicting crop responses to irrigation inputs within the MPC framework [33,50]. Variability in environmental conditions and biological factors introduces uncertainties in model parameters and predictions, which can undermine the effectiveness of irrigation scheduling. Managing and mitigating these uncertainties pose significant challenges, as they can affect the reliability and accuracy of MPC-based irrigation decisions.

Computational Burden: The computational demands associated with solving complex optimization problems in real-time present a notable challenge for the implementation of MPC in irrigation scheduling [30,51]. MPC requires iterative solutions to optimization problems to determine optimal irrigation decisions over specified time horizons. However, the computational burden of these tasks can be substantial, especially in large-scale irrigation systems or systems with high-frequency control. Consequently, addressing this challenge requires the development of efficient algorithms and computational strategies to optimize irrigation schedules effectively.

Implementation Costs: The adoption of the MPC framework in irrigation scheduling entails significant investment in technology, including sensors, actuators, and control systems [52]. These technological requirements contribute to the initial setup costs and ongoing maintenance expenses associated with MPC implementation. For smallholder farmers or resource-limited agricultural enterprises, the financial burden of implementing and maintaining MPC systems may present a formidable barrier to adoption, limiting their accessibility and widespread use.

Operator Training and Expertise: Effectively utilizing MPC systems for irrigation scheduling requires operators to possess the necessary training and expertise in interpreting model outputs and making informed irrigation decisions [53]. However, providing adequate training and support to operators can be challenging, particularly in settings with limited access to technical expertise. Ensuring that operators have the requisite knowledge and skills to navigate MPC systems effectively is essential for maximizing their benefits in agricultural water management.

Adaptation to Dynamic Conditions: MPC systems must be capable of adapting to dynamic changes in environmental conditions, such as fluctuations in weather patterns or sudden shifts in crop water requirements [54]. However, ensuring that MPC frameworks can respond appropriately to these changes in real-time without compromising performance presents a significant challenge. Developing adaptive algorithms and control strategies that enable MPC systems to adjust irrigation schedules dynamically is crucial for enhancing their effectiveness and robustness in addressing evolving agricultural water management needs.

5. Conclusion

In conclusion, this article has demonstrated the feasibility and effectiveness of using a model predictive controller (MPC) for real-time irrigation scheduling. The proposed MPC algorithm considers important variables such as weather forecasts and soil moisture levels and is capable of optimizing irrigation schedules in real-time to achieve efficient and effective water management. The results of the simulation studies show that the MPC approach can significantly improve irrigation efficiency and reduce water waste, compared to traditional scheduling methods. These findings highlight the potential for the MPC approach to play a crucial role in the development of sustainable and effective irrigation systems, and in addressing the challenges of managing water resources in agriculture. Further research is needed to validate the proposed approach in real-world settings, and to explore its scalability and applicability in other agricultural domains.

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Data availability statement

Data will be made available through the corresponding author.

CRediT authorship contribution statement

Erion Bwambale: Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Felix K. Abagale:** Writing – original draft, Validation, Supervision, Resources, Methodology. **Geophrey K. Anornu:** Writing – review & editing, Supervision, Project administration, Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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